



Arrays

Practical Skills Module 2: 10

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Practical Skills Module 2: Topic 10

Topic Code	PM-02-PS010
Topic	Create Arrays in Python

Scope of Practical Skill

Given an applicable instruction and access to a learning platform, the learner must be able to:

- PA1001 Create arrays
- PA1002 Import arrays
- PA1003 Access array elements
- PA1004 Insert elements
- PA1005 Modify elements
- PA1006 Pop an element from array
- PA1007 Delete elements
- PA1008 Search and get the index of a value in an array
- PA1009 Reverse an array
- PA1010 Count the occurrence of a value in array
- PA1011 Traverse an array

Applied Knowledge

- AK1001 Concept, definition and functions of Python programming functionalities

Internal Assessment Criteria

- IAC1001 Expected results with Python arrays are achieved

Resources:

Knowledge	Information from Python Programmer Curriculum
National Curriculum Framework	900221-000-00-PM-02



Practical task

Follow the facilitator's instructions to complete the following activities:

Scenario: Inventory Management for an Electronics Store

An electronics store manager uses Python to manage an inventory of products. The inventory is stored in arrays, where each array represents a category of products (e.g., laptops, phones, tablets). The manager needs to perform operations like adding new products, modifying existing ones, deleting discontinued items, searching for specific products, and analysing the inventory.

Create a Python file called **inventory_manager.py** and complete the steps below:

Step 1: Create arrays (lists) for inventory categories.

- Create two arrays (lists) to store the provided laptop and phone brands

```
laptops : ["Dell", "HP", "Lenovo", "Apple"]
```

```
phones : ["Samsung", "Apple", "OnePlus", "Nokia"]
```

Step 2: Access array elements.

- Access and print the **first element** in the `laptops` list.

Step 3: Insert a new product.

- Insert "Acer" into the `laptops` list at **position 2**.
- Print the updated list to confirm the change.

Step 4: Modify an existing product.

- Change the **second item** in the `phones` list (index 1) to "iPhone".
- Print the updated `phones` list.
 - *Expected output:* Updated phones list: ['Samsung', 'iPhone', 'OnePlus', 'Nokia']

Step 6: Pop an element from an array.

- Remove the **last item** from the `phones` list using the `pop()` method.
- Print the list after the item is removed.

Step 7: Delete an element at a specific position.

- Remove the item at index 1 from the `phones` list.
- Print the updated list to confirm the deletion.

Step 8: Search and get the index of a value.

- Search for the product "Lenovo" in the `laptops` list.
- Find and print its index.
 - *Expected output: Index of Lenovo: 2*

Step 9: Reverse the array.

- Reverse the order of elements in the `phones` list.
- Print the list after reversing it.

Step 10: Count how many times a value appears.

- Count how many times "Apple" appears in the `phones` list.
- Print the result.
 - *Expected output: Apple appears 1 time(s) in phones*

Step 11: Traverse the array.

- Use a for loop to **traverse and print** each product in the `laptops` list.
- Label the output for clarity.
 - *Example output:*
 Laptop: Dell
 Laptop: HP
 Laptop: Lenovo
 Laptop: Apple



Take note

Important: Be sure to upload all files required for the task submission inside your task folder and then click "Request review" on your dashboard.
